Adam optimizer

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1 General set up

1.1 Adam

Definition 1.1 (Innovation). *Let $\mathcal{U}$ be a measurable space. A pair (X, U) consisting of*

(i) a $\mathcal{U}$-valued random variable U and

(ii) a product measurable mapping $X\colon \mathcal{U} \times \mathbb{R}^d \to \mathbb{R}^d$

is called innovation.

$\mathcal{U}$ is the random samples. And X puts $\mathcal{U}$ into stochastic gradient.

Definition 1.2 (Adam optimizer)**.** *Let*

(i) $\alpha, \beta \in [0,1)$, $\epsilon \in (0,\infty)$ *with* $\alpha < \sqrt{\beta}$ *(the* damping parameters*),*

(ii) $n_0 \in \mathbb{N}_0$, $\theta_{n_0}, m_{n_0} \in \mathbb{R}^d$, $v_{n_0} \in [0,\infty)^d$ *(the* initialisation*),*

(iii) a decreasing $(0,\infty)$*-valued sequence* $(\gamma_n)_{n\in\mathbb{N}}$ *(the* sequence of step-sizes*) and*

(iv) (X, U) *an innovation.*

An $\mathbb{R}^d$*-valued stochastic process* $(\theta_n)_{n\in\mathbb{N}_0\cap[n_0,\infty)}$ *is called* Adam algorithm with damping parameters (α,β,ϵ) and step-sizes (γ_n) started at time n_0 in $(\theta_{n_0}, m_{n_0}, v_{n_0})$*, if it satisfies for every* $n \in \mathbb{N}\cap(n_0,\infty)$ *and* $i = 1,\dots,d$

$$\theta_n^{(i)} = \theta_{n-1}^{(i)} + \gamma_n \, \sigma_n^{(i)} \, m_n^{(i)}, \tag{1}$$

where $(U_n)_{n\in\mathbb{N}\cap(n_0,\infty)}$ *is a family of independent copies of* U *and for* $n \in \mathbb{N}\cap(n_0,\infty)$ *and* $i = 1,\dots,d$*,*

- $m_n = \alpha \, m_{n-1} + (1-\alpha)\, X(U_n, \theta_{n-1})$,
- $v_n^{(i)} = \beta \, v_{n-1}^{(i)} + (1-\beta)\,(X^{(i)}(U_n,\theta_{n-1}))^2$ *and*
- $\sigma_n^{(i)} = \dfrac{1}{\sqrt{v_n^{(i)}/(1-\beta^n)} + \epsilon}$.

We call the sequence $(t_n)_{n\in\mathbb{N}_0}$ *given by*

$$t_n = \sum_{k=1}^{n} \gamma_k \tag{2}$$

the training times *of the Adam algorithm.*

2 Adam sequance space

Consider a sequence of history:

$$\mathbf{x} = (x_k)_{k\in-\mathbb{N}_0} = (x_0, x_{-1}, x_{-2}, \cdots)$$

Consider an operator: $\boxed{\text{The whole history sequence}} \Rightarrow \boxed{\text{Adam direction}}$

Definition 2.1. *Let* $\alpha, \beta, \epsilon \in [0,\infty)$ *with* $\alpha < \sqrt{\beta} < 1$ *and let* $(\varrho_k)_{k\in-\mathbb{N}_0}$ *as in* (4)*.*

$$g\colon \ell_\varrho^d \to \mathbb{R}^d, \qquad \mathbf{x} = (x_k)_{k\in-\mathbb{N}_0} \mapsto \left(\frac{(1-\alpha)\sum_{k\in-\mathbb{N}_0} \alpha^{-k} x_k^{(i)}}{\epsilon + \sqrt{(1-\beta)\sum_{k\in-\mathbb{N}_0} \beta^{-k} (x_k^{(i)})^2}} \right)_{i\in\{1,\dots,d\}} \tag{3}$$

One more natural way is:

$$g(\mathbf{x}) = \frac{m(\mathbf{x})}{\sqrt{v(\mathbf{x})} + \epsilon}$$

We are curious about how *senstive* g is to the change of x_k in $\mathbf{x}$.
One natural idea is to find the partial derivative of g:

Definition 2.2 (Adam sequence space). *Let $\alpha, \beta \in [0,1)$, $\epsilon \in (0,\infty)$ with $\alpha < \sqrt{\beta}$ be damping factors. We let*

$$\varrho_k = (1-\alpha)\epsilon^{-1}\Big(\alpha^{-k} + \frac{1}{\sqrt{1-\alpha^2/\beta}}\beta^{-k/2}\Big), \qquad \text{for all } k \in -\mathbb{N}_0, \tag{4}$$

and denote by ℓ_ϱ^d the space of all sequences $\mathbf{x} = (x_k)_{k\in-\mathbb{N}_0} \in (\mathbb{R}^d)^{-\mathbb{N}_0}$ with

$$\|\mathbf{x}\|_{\ell_\varrho^d} \coloneqq \sum_{k\in-\mathbb{N}_0} \rho_k\,|x_k| < \infty. \tag{5}$$

We equip the space with the norm $\|\cdot\|_{\ell_\varrho^d}$. Moreover, we let for a tuple $(m,v) \in \mathbb{R}^d \times [0,\infty)^d$

$$\begin{aligned}(m,v)_{\ell_\varrho^d} = \inf\Big\{\|\mathbf{x}\|_{\ell_\varrho^d} : \mathbf{x}\in\ell_\varrho^d,\ m = (1-\alpha)\sum_{k\in-\mathbb{N}_0}\alpha^{-k}x_k \quad \text{and}\\ v^{(i)} = (1-\beta)\sum_{k\in-\mathbb{N}_0}\beta^{-k}(x_k^{(i)})^2 \text{ for } i = 1,\dots,d\Big\},\end{aligned} \tag{6}$$

where the infimum of the empty set is ∞.

(see Theorem 2.3 below). First, we note that g is well defined since for every $\mathbf{x} \in \ell_\varrho^d$ and $i \in \{1,\dots,d\}$ we have that the series in the nominator in (3) is summable. The denominator in (3) may attain the value ∞ in which case g is defined to be zero.

Lemma 2.3. *Let $\alpha, \beta, \epsilon \in [0,\infty)$ with $\alpha < \sqrt{\beta} < 1$ and let $(\varrho_k)_{k\in-\mathbb{N}_0}$ and g as defined in (4) and (3), respectively. Then g is Lipschitz continuous with Lipschitz constant 1 and g is uniformly bounded by $\sqrt{d}\frac{1-\alpha}{\sqrt{1-\beta}}\frac{1}{\sqrt{1-\alpha^2/\beta}}$.*